

AMPERE'S LAW

Q. 1 State and explain Biot – Savart law?

BIOT-SAVART LAW

The magnitude of magnetic field at a point due to current distribution is directly proportional to amount of current and effective length of conductor and inversely proportional to square of distance between point and current distribution.

EXPLANATION

The magnetic field is generated around a conductor when current flows through it. Now place another current carrying conductor in the magnetic field of first conductor. The magnetic force acts on second conductor due to interaction of magnetic field of both conductors. Similarly first conductor also experiences magnetic force which is placed in magnetic field of second conductor.

MATHEMATICAL TREATMENT

Now consider two conductors through which current I_1 and I_2 flows in opposite direction as shown in fig. Divide the first wire into differential element dS_1 and

then define for each element a length vector dS_1 that has magnitude length dS_1 and direction is the direction of current in dS_1 . In this way, differential current length element of first wire is $I_1 dS_1$. Similarly differential current length element of second wire is $I_2 dS_2$.

The magnetic force dF_{12} exerted on current element $I_2 dS_2$ which is placed in magnetic field B_1 of first conductor is

$$dF_{12} = 12 (dS_2 \times)$$

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

The magnetic force dF_{21} exerted on current, element $I dS_1$ which is Placed In magnetic field B_2 of second conductor is $dF_{21} = (dS_1 \times B_2)$

Take a point P having distance r from charge dq of some charge distribution. The electric field dE at point P due to charge dq is given as

The direction of electric field is given by electric lines of force which come out from positive charge and enters into negative charge

Similarly take a point P having distance r from current element $I dS$ of some currents distribution. The magnetic field dB at point P due to current element $I dS$ is given as

$$\frac{\mu_0 I dS \sin \theta}{4\pi r^2}$$

$$\frac{\mu_0 I dS \times \hat{r}}{4\pi r^2}$$

Where $\hat{r}$ is unit vector directed from $I dS$ to point P and μ_0 is proportionality constant having value $4\pi \times 10^{-7} \text{ Tm/A}$. The constant is called permeability of free space having value $4\pi \times 10^{-7} \text{ Tm/A}$.

The direction of magnetic field is the same as cross product of $dS \times \hat{r}$. According to right hand rule direction of magnetic field is into plane of paper.

The magnitude of magnetic field is given as

$$dB = \frac{\mu_0 I dS \sin \theta}{4\pi r^2}$$

$$\frac{\mu_0 I dS \sin \theta}{4\pi r^2}$$

Where θ is the angle between dS and r . The net magnitude of magnetic field due to entire current distribution is

$$B = \int dB$$

CHAPTER 9]

249

$$B = \int \frac{\mu_0}{4\pi} \frac{I dS \sin\theta}{r^2}$$

The net magnetic field in vector form due to entire current distribution is



$$\vec{B} = \int d\vec{B}$$

$$\vec{B} = \int \frac{\mu_0}{4\pi} \frac{I d\vec{S} \times \hat{r}}{r^2}$$

$$\vec{B} = \int \frac{\mu_0}{4\pi} \frac{I d\vec{S} \times \vec{r}}{r^3} \quad [\vec{r} = r \hat{r}]$$

Q.2 By using Biot-Savart law, determine the magnetic field at a point having distance R away from the long straight current carrying wire.

MAGNETIC FIELD DUE TO STRAIGHT WIRE

The magnetic field is generated when current flows through a straight conductor. Consider a long straight wire of infinite length through which current I flows. Take a point P having perpendicular distance R from wire. The magnitude of magnetic field dB at point P due to current element $I dS$ of wire having distance r is given by Biot-Savart law as

$$dB = \frac{\mu_0}{4\pi} \frac{I dS \sin\theta}{r^2}$$

From fig

$$r^2 = R^2 + S^2 \text{ and}$$

$$\sin\theta = \frac{R}{r} = \frac{R}{\sqrt{R^2 + S^2}}$$

Put these value in above eq

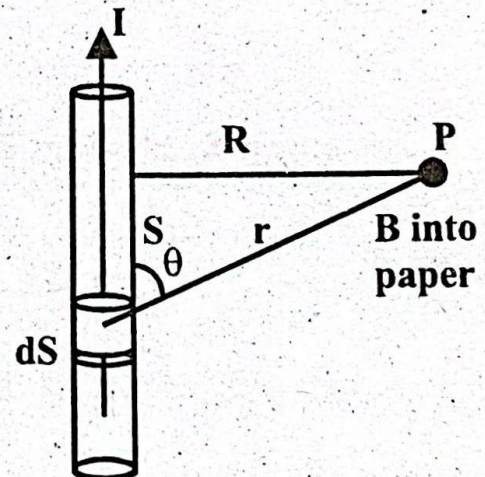
$$dB = \frac{\mu_0}{4\pi} \left(\frac{I dS}{R^2 + S^2} \right) \left(\frac{R}{\sqrt{R^2 + S^2}} \right)$$

$$dB = \frac{\mu_0 I R}{4\pi} \frac{dS}{(R^2 + S^2)^{3/2}}$$

The net magnetic field is

$$B = \int_{-\infty}^{\infty} dB = \int_{-\infty}^0 dB + \int_0^{\infty} dB = \int_0^{\infty} dB + \int_0^{\infty} dB$$

$$B = 2 \int_0^{\infty} dB = 2 \int_0^{\infty} \frac{\mu_0 I R}{4\pi} \frac{dS}{(R^2 + S^2)^{3/2}}$$



When S = 0	When S = ∞	Put S = R cot θ Then (1/S - R cos θ) dθ
cot θ	- R	cot θ
$\frac{\cos \theta}{\sin \theta}$	Sin θ	
Sin θ	Cos θ	
cos θ	sin θ	
cos - 0 = 90°		

$$R \csc^2 \theta d\theta$$

$$2 \int_{90^\circ}^0 (R^2 + R^2 \cot^2 \theta)^{3/2} d\theta$$

i Egg-Q-dL

$$\frac{2 \pi R^2 \csc^2 \theta d\theta}{(I \cot^2 \theta)^{3/2}}$$

$$2 \pi R^2 \csc^2 \theta d\theta (I \cot^2 \theta)^{3/2}$$

$$\csc^2 \theta d\theta$$

$$\csc^2 \theta d\theta$$

$$\int \sin \theta d\theta$$

$$1 - \cos \theta$$

$$(1 - \cos 90^\circ - \cos \theta)$$

This is the magnitude of magnetic field at a point P having perpendicular distance R from current carrying straight conductor. The direction of magnetic field-is the same as the direction of $d\mathbf{S} \times \mathbf{r}$. It is directed into plane of paper according to right hand rule.

Q.3 By using Biot—Savart law, derive an expression of magnitude Of magnetic field on axis perpendicular to current carrying loop and at center or loop?

MAGNETIC FIELD ON CIRCULAR CONDUCTING LOOP

Consider a conducting loop of radius R through which Current I flows in anticlockwise direction as shown in fig, Take a point P on axis of loop having distance z from the plane of loop where we want. to evaluate magnetic field generated due to flow Of current in loop.

Take a small current element IdS of loop at position M having distance r from point P. The

angle θ between current element IdS and r is 90° . The magnetic field dB at P will be perpendicular to

plane defined by IdS and r .

Its magnitude is given by Biot—Savart law as

$$\frac{I dS \sin 90^\circ}{r^3} \quad (1)$$

IdS

Now resolve the magnetic field dB into rectangular components $dB \cos \theta$ along the axis of the loop and $dB \sin \theta$ right angle to axis of the loop. $IdS \cos \theta - dB \cos \theta$

$dB \sin \theta$ similarly take an other current element IdS at location N such that line MN is a diameter of the loop. The magnitude of magnetic field due to this current element is given by Biot—Savart law as

$$dB_2 = \frac{I dS \sin 90^\circ}{r^3} \quad (2)$$

Resolve the magnetic field dB_2 due to this element into rectangular

components $dB_2 \cos \theta$ $dB_2 \sin \theta$ $dB_2 \cos \theta = dB_2 \sin \theta$

Comparing eq(1) and eq(2) dB_2

The net magnetic field right angle to axis of the loop

is $dB \cos \theta - (dB \sin \theta) + (dB_2 \sin \theta)$

$= -dB \sin \theta + dB_2 \sin \theta$

$dB \cos \theta - dB \sin \theta + dB \sin \theta$

,The net magnetic field along the axis of loop is dB

$- (dB \sin \theta) + (dB_2 \sin \theta)$

$dB \cos \theta + dB_2 \cos \theta$

$dB \cos \theta + dB_2 \cos \theta$

ELECTRICITY AND MAGNETISM (11th class)

The net magnetic field due to both current elements

$dB = (dB \cos \theta)^2 + (dB \sin \theta)^2$

$dB \cos \theta + dB \sin \theta$

$$dB = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}} dz$$

$$dB = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}} dz$$

The net magnetic field to entire current loop is

$$B = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

$$2\pi R^2 (R^2 + z^2)^{3/2}$$

$$2\pi R^2 (R^2 + z^2)^{3/2} \text{ OR}$$

Where from fig. $r^2 = R^2 + Z^2$ and $J dS = \frac{I}{2\pi R} (2\pi R dz)$ because current element is taken at both ends of diameter.

$$\frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

This is the magnitude of magnetic field at a point P on axis of the loop having distance z from the plane of loop. To calculate magnetic field at the center of loop put

$$\frac{\mu_0 I R^2}{2R^3}$$

This is the magnitude of magnetic field at the center of the loop. If $z \gg R$, then R is zero as compared to z in term $(R^2 + z^2)$

The magnetic field when loop has N turns is

$$B = \frac{\mu_0 N I R^2}{2(R^2 + z^2)^{3/2}}$$

The magnetic field when A is the area of loop

$$B = \frac{\mu_0 N I A}{2 \pi r^2}$$

Where $\mu = \mu_0 \mu_r$ is called magnetic dipole moment.

Two long parallel wires carrying currents I_1 and I_2 are separated by a distance d . Find magnitude of magnetic force experienced by each wire. Alternatively, show that parallel currents attract and anti-parallel currents repel each other.

PARALLEL CURRENTS ATTRACT

The parallel currents flowing through conductors attract each other. Consider two parallel conductors P and Q each of length L having separation d . The current I_1 and I_2 flows through these conductors in the same direction.

The magnitude of magnetic field B_1 generated by conductor P at a location where conductor Q is placed is given by Biot-Savart law

$$\frac{\mu_0 I_1 L}{2 \pi d}$$

The direction of magnetic field B_1 is into plane of paper according to right hand rule.

The conductor Q having length L is now immersed in magnetic field B_1 . The magnetic force experienced by conductor Q having current I_2 is

into
plane
of
paper

Put the value of B_1

$$F_{21} = I_2 L B_1$$

$$(1)$$

$$F_{21} = \frac{\mu_0 I_1 I_2 L^2}{2 \pi d}$$

This is magnitude of magnetic force which is acting on conductor Q . The direction of this magnetic force $F_{21} = I_2 (L \times B_1)$ is towards conductor P according to right hand rule.

Similarly magnitude of magnetic field generated by conductor Q carrying current I_2 at a location where conductor P is placed is given by Biot-Savart law as

$$\frac{\mu_0 I_2}{2\pi d}$$

ELECTRICITY AND MAGNETISM (B.Sc. Classes)

The direction of magnetic field B_2 is out of plane of paper according to right hand rule. The conductor P having length L is now immersed in magnetic field B_2 . The magnetic force experienced by conductor P carrying current I_1 is

$$F_{12}$$

Put the value of B_2

$$F_{12} = I_1 L \frac{\mu_0 I_2}{2\pi d} \quad (2)$$

This is magnitude of magnetic force which is acting on conductor P, The direction of this magnetic force is $F_{12} = I_1 (L \times B_2)$ towards conductor Q according to right hand rule.

Comparing eq(i) and eq(2)

$$F_{12} = F_{21}$$

It shows that F_{12} and F_{21} are equal in magnitude but acting towards each other and their point of application is not same. Therefore, it is proved that parallel currents attract each other.

ANTIPARALLEL CURRENT REPELS

The antiparallel currents flowing through conductors repel each other. Consider two parallel conductors P and Q each of length L having separation d . The current I_1 and I_2 flows through these conductors in the opposite direction.

The magnitude of magnetic field B_1 generated by conductor P at a location where conductor Q is placed is B_1 , given by Biot-Savart law as into plane of paper

the direction of magnetic field B_1 is into plane of paper according to right hand rule.

The conductor Q having length L is now immersed

into
plane

in magnetic field B_1 . The magnetic force experienced by conductor Q having current I_2 is

of
paper

$$F_{21} = I_2 L$$

Put the value of B_1

$$= \frac{\mu_0 I_1}{2\pi d}$$

2nd

(1)

This is magnitude of magnetic force which is acting on conductor Q. The direction of this magnetic force $\vec{F}_{21} = I_2 (\vec{L} \times \vec{B}_1)$ is away from conductor P according to right hand rule.

Similarly magnitude of magnetic field generated by conductor Q carrying current I_2 at a location where conductor P is placed is given by Biot-Savart law as

$$B_2 = \frac{\mu_0 I_2}{2\pi d}$$

The direction of magnetic field B_2 is into plane of paper according to right hand rule. The conductor P having length L is now immersed in magnetic field B_2 . The magnetic force experienced by conductor P carrying current I_1 is

Put the value of B_2

$$F_{12} =$$

2nd

(2)

This is magnitude of magnetic force which is acting on conductor P. The direction of this magnetic force is $\vec{F}_{12} = I_1 (\vec{L} \times \vec{B}_2)$ away from conductor Q according to right hand rule.

Comparing eq(1) and eq(2)

$$F_{21}$$

It shows that F_{12} and F_{21} are equal in magnitude but acting away from each other and their point of application is not same. Therefore, it is proved that antiparallel currents repel each other.

Q. 5 State and explain Ampere law, write its integral and differential form.

AMPERE LAW

The Ampere law states that magnetic field is generated around a conductor when current flows through it. The magnitude of generated magnetic field at a point is directly proportional to amount of current and inversely proportional to distance between point and straight current carrying conductor.

DIRECTION OF MAGNETIC FIELD

The direction of magnetic field is given by right hand rule stated as

Grasp the current carrying wire in your right hand in such a way that erect thumb is along the direction of current. The curled fingertips around the wire indicates the direction of magnetic field.

EXPLANATION

Consider a straight conductor through which current I flows. The magnetic field B is generated around it in the form of circles. The magnitude of magnetic field B depends upon current I and distance r between conductor and point P where we want to evaluate magnetic field.

Where μ_0 is constant of proportionality and μ_0 is called permeability of free space

Its value is $4\pi \times 10^{-7} \text{ Wb/A}\cdot\text{m}$

Where $2\pi r$ is the length of path called circumference of circle and denoted by

The value of B is constant around the path

$$B \cdot dl = \mu_0 I$$

$$B \int dl \cos 0^\circ = \mu_0 I$$

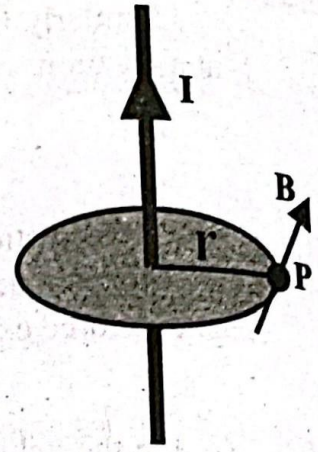
$$B \propto I$$

$$B \propto \frac{1}{r}$$

Combining both experimental facts

$$B \propto \frac{I}{r}$$

$$B = \frac{\mu_0}{2\pi} \left(\frac{I}{r} \right)$$



Now consider an amperian loop to find magnetic field set up by current in a long straight wire such that wire is perpendicular to plane of loop and current flows out of wire as shown in fig.

Take a small length element dr of the amperian loop.
The total line integral for amperian loop is

$$B \cdot dr = \oint B \cdot dr = \mu_0 I$$

$$B dr \cos 0$$

AMPERE'S CIRCUITAL LAW

For any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop.

INTEGRAL FORM OF AMPERE'S LAW

$$\oint B \cdot dr = \mu_0 I$$

91 AMPERE'S LAW

The value of current in terms of current density is

$$I = \oint_C \mathbf{J} \cdot d\mathbf{a}$$

The above eq becomes

This is called, integral form of Ampere's law

DIFFERENTIAL FORM

The integral form of Ampere's law is

$$\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 \oint_S \mathbf{J} \cdot d\mathbf{a} \quad (1)$$

The Stoke's theorem is

$$(2)$$

Comparing eq (1) and eq(2)

$$\oint_C \mathbf{J} \cdot d\mathbf{a}$$

$$\oint_C (\nabla \times \mathbf{B} - \mu_0 \mathbf{J}) \cdot d\mathbf{a} = 0$$

Since $d\mathbf{a}$ cannot be zero so

This is called differential form of Ampere's law.

Q.-6 Apply Ampere's law to calculate magnetic field within and outside a current carrying solenoid?

SOLENOID

A long tightly wound cylindrical conducting wire is called solenoid. The magnetic field is generated when current flows through windings of solenoid. It provides uniform magnetic field and used to store magnetic energy.

The magnetic field inside solenoid is uniform and strong called strong magnetic field. The magnetic field outside the solenoid is weak.

The iron core solenoid has much greater magnetic field as compared to the air core solenoid.

MAGNETIC FIELD INSIDE SOLENOID Consider a solenoid having N turns and length C . The Current I flows through each turn of solenoid. The turns per unit length of solenoid are $n = \frac{N}{C}$ — The strong magnetic field

B is generated inside solenoid which is directed along axis of solenoid. The magnetic field outside solenoid is so weak that it is considered zero. Detail of bottom of rectangle inside solenoid

$B_{out} =$

Take amperian loop in the shape of rectangle to calculate magnetic field inside solenoid by applying Ampere's law

$$\oint \mathbf{B} \cdot d\mathbf{r}$$

$d\mathbf{r}$

$$\int \mathbf{B} d\mathbf{r} \cos 0^\circ + \int \mathbf{B} d\mathbf{r} \cos 90^\circ + \int \mathbf{B}_{out} d\mathbf{r} \cos 180^\circ + \int \mathbf{B} d\mathbf{r} \cos 90^\circ =$$

$$\int \mathbf{B} d\mathbf{r} (0) + J(O) d\mathbf{r} (-l) + \int \mathbf{B} d\mathbf{r} (0) -$$

The length of loop from a to b is h so $d\mathbf{r} = h$

Unit length of solenoid has turns n

h length of loop intercept turns nh

TOTAL CURRENT IN TURNS OF SOLENOID

One turn of solenoid has current — 10 nh
 turns of solenoid have current =

$nhlo$

$nhlo$

91 AMPERE'S LAW

Put eq (2) in eq (i)

$$Bh - (nhl())$$

$$B = n 10$$

This is the magnitude of magnetic field inside solcnoid.

DIRECTION OF MAGNETIC FIELD

The direciion of magnetic field is given by right hand rule stated 'as "hold the solenoid •in your right hand in such a way that fingertips are along direction of current then erect thumb indicates the direction of magnetic field which along axis of solenoid.

The solenoid acts as bar magnet. The thumb side IS north pole and other side is south pole. The magnetic field lines come out from north pole and enter into south pole.

MAGNETIC FIELD OUTSIDE SOLENOID

The solenoid acts as bar magnet when current flows through the windings of solenoid. The magnetic field lines come out from north pole and enter into south pole. field lines are concentrated inside the solenoid while spread in space outside the solenoid. That is why magnetic field outside solenoid becomes weak.

Consider a solenoid having n turns per unit length. The current 10 flows through each turn of solenoid. To calculate magnetic field outside solenoid, imagine an amperian loop having radius r which encloses one turn. The magnetic field lines will be tangential.

The Ampere's law is

$$\oint \mathbf{B} \cdot d\mathbf{r} = \mu_0 I$$

$$\oint B \cos 0^\circ dr = \mu_0 I$$

$$B \int dr = \mu_0 I$$

$$B (2\pi r) = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

Amperian-loop

(1)

This is magnetic field outside the solenoid. The magnetic field inside the solenoid is

$$B = \frac{\mu_0 n I}{2} \quad (2)$$

Divide eq(1) and eq(2)

$$\frac{B}{B} = \frac{\mu_0 n I}{2 \mu_0 n I} \quad \text{on } 10$$

$$2 \mu_0 n I$$

It shows B_t is smaller than B because $2 \mu_0 n I$ is very 'large factor. That is why magnetic field outside solenoid is Weaker than inside.



AND

Q. 7 Apply Ampere's law calculate magnetic field at the interior points of a current carrying toroid.

TOROID

The solenoid bent into circular shape is called toroid. A strong magnetic field is generated when current flows through windings of a solenoid. These magnetic field lines are along the axis of the solenoid. These magnetic field lines become concentric circles in toroid.

The toroid is a very useful device used in tape heads and tokamaks. It is a central feature of the tokamak which is the basis of fusion power reactor.

MAGNETIC FIELD DUE TO TOROID

The hollow space of toroid has no magnetic field. There is no magnetic field around the windings of toroid because magnetic field generated between two adjacent turns is cancelled out. Therefore, magnetic field outside toroid is zero.

All the magnetic field is present inside the tubular path of toroid. The magnetic field is not constant over the entire cross section of the toroid.

Consider a toroid having N turns and each turn carries current I . The total current is given as

$$I_{\text{total}} = NI$$

To calculate magnetic field B at a point P inside toroid, imagine an amperian loop of radius r such that point P lies on it. The magnetic field B and small length element dr of amperian loop are along tangent direction so makes zero angle.

Apply Ampere's law

$$\oint B \cdot dr = \mu_0 NI$$

$$B(2\pi r) = \mu_0 NI$$

This is magnitude of magnetic field inside toroid. It depends upon turns of toroid and radius r .

DIRECTION OF MAGNETIC FIELD

, The direction of magnetic field within toroid is given by right hand rule stated as "curl the fingers of your right hand in the direction of the current. then extended thumb indicates the direction of the magnetic field". These field lines are circular.

SHORT QUESTIONS

Q.1 Is wire B carrying uniform current. for all points within a circular loop of

Yes, magnetic field is uniform for all points within a
 Ans: circular loop of wire carrying current because field lines are
 concentrated within loop.

Is there any way to set up a magnetic field other than by causing charge to move.
 The origin of magnetism is the motion of electrons. Therefore, there IS no any other
 way to set up a magnetic field except by causing charge to move. If you change
 electric field, you produce magnetic field.

What is angle between direction of flow of current and external magnetic field to
 have magnetic force of maximum value.

Ans: The angle between direction of flow of current and external magnetic field to have
 magnetic force of maximum value is 90°

What type of magnetic field exist around a current
 carrying straight conductor.

Ans. • The magnetic field around current carrying straight
 conductor is in the form of concentric circles.

Q:5 Can the path of integration around which we apply
 Ampere's law pass through the conductor.

Ans.' The path of integration around which we apply
 Ampere's law does not pass through the conductor.
 The path is always around the conductor.

Q:6 In electronics, wires that carry equal and opposite currents are often twisted together to reduce their magnetic effect at distant points. Why is this effective?

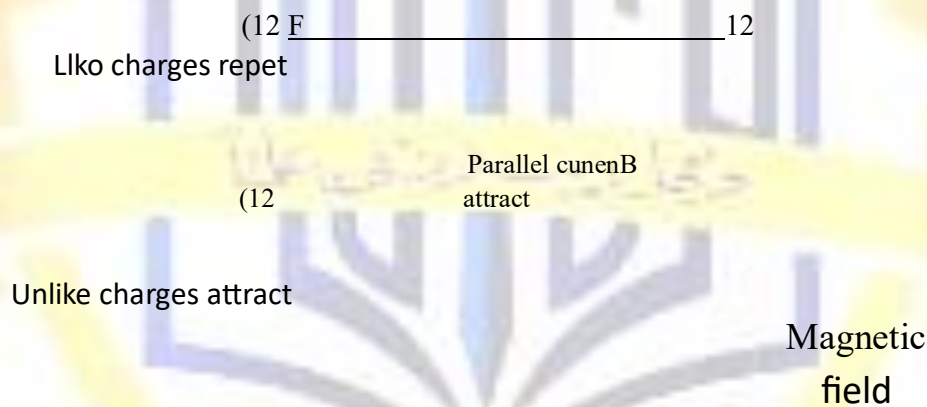
Ans: In electronics, wires that carry equal and opposite currents are often twisted together to reduce their magnetic effect at distant points because back emf is not produced so original resistance is unaffected.

Q:7 What is the general rule for deciding about Coulomb force between like charges and between parallel currents.

Ans: The like Charges repel each other and unlike charges attract each other with a force called Coulomb force.

The parallel currents through conductors attract and anti-parallel current through conductors repel each other with a force.

Classes)



Q:8 A beam of 20 MeV protons emerges from a cyclotron. Do these particles cause a magnetic field?

Ans: A beam of 20 MeV protons emerges from a cyclotron give rise a magnetic field because moving charges generate magnetic field.

Q:9 Consider a magnetic field line. Is the magnitude of B constant or variable along a line?

Ans: The magnitude of B is variable along a line because B is inversely proportional to distance.

Q:10 A circular loop of wire lies on the floor of the room in which you are sitting, It carries a constant current I in clockwise direction as viewed from above. What is the direction Of magnetic dipole moment of this current loop?

Ans: According to right hand rule, the direction of the magnetic dipole moment of current loop through which current flows is towards floor of the room.

Q:11 A current is sent through a vertical spring from whose lower end a weight is hanging. What will happen?

Ans: The spring starts oscillations, when current is sent through a vertical spring from whose lower end a weight is hanging because magnetic field is generated due to current which give rise to force.

Q:12 Show that the unit of the expression $\frac{1}{I L}$ comes out to be Joule/amp^2

NBA Ans.

$\frac{1}{I L}$
 $\frac{\text{newton meter}}{\text{ampere}}$

$\frac{1}{I L}$
 $\frac{\text{Joule}}{\text{ampere}^2}$

13 Discuss analogies and difference between Coulomb's law and Biot—Savart law?

Ans. Coulomb's law gives electric force between two charges. The Biot—Savart law gives magnetic force between two currents.

The Biot—Savart law gives magnetic

force between two charges

force between two currents

The electric field dE at point P due to charge dq according to Coulomb's law is given as $\frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$. The magnetic field dB at point P due to current element IdS according to Biot—Savart law is given as $\frac{\mu_0}{4\pi} \frac{IdS \times \vec{r}}{r^3}$.

Biot — Savart law is given as

$\frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$

$$dB = \frac{\mu_0}{4\pi} \frac{IdS \times \vec{r}}{r^3}$$

Direction of electric field is given by lines of force which come out from positive charge and enter into negative charge. The direction of magnetic field is the direction of cross product of $dS \times \vec{r}$.

proportionality constant in Coulomb's law is $k = \frac{1}{4\pi\epsilon_0}$. The proportionality constant in Biot—Savart law is $k = \frac{\mu_0}{4\pi}$.

Where ϵ_0 is called permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N m}^2$. Where μ_0 is called permeability of free space having value $4\pi \times 10^{-7} \text{ T m/A}$.

The like charges repel each other and unlike charges attract each other with a Coulomb force. The parallel currents through conductors attract and anti-parallel current through conductors repel each other with a magnetic force.

SAMPLE PROBLEMS

P:] In the Bohr model of hydrogen atom, the electron circulates around the nucleus in path $5.29 \times 10^{-11} \text{ m}$ at frequency $6.6 \times 10^{15} \text{ Hz}$; (a) What is the value of B at centre of orbit. (b) What is magnetic dipole moment?

Sol.

DATA $q = 1.6 \times 10^{-19} \text{ C}$

$$9.1 \times 10^{31} \text{ kg}$$

$$5.29 \times 10^{11} \text{ m}$$

$$6.6 \times 10^{15} \text{ Hz}$$



CALCULATIONS

$$I = \frac{q}{t} = qf$$

$$I = 1.6 \times 10^{-19} \times 6.6 \times 10^{15} = 1.05 \times 10^{-3} \text{ A}$$

$$(a) \quad B = \frac{\mu_0 I}{2r}$$

$$B = \frac{4\pi \times 10^{-7} \times 1.05 \times 10^{-3}}{2 \times 5.29 \times 10^{-11}} = 13 \text{ T}$$

$$(b) \quad \mu = I A$$

$$\mu = I \pi r^2$$

$$\mu = 1.05 \times 10^{-3} \times \pi (5.29 \times 10^{-11})^2$$

$$\mu = 9.2 \times 10^{-24} \text{ A} \cdot \text{m}^2$$

P:2 A solenoid has length 1.23 m and inner diameter 3.55 cm. It has 5 layers of windings and having 850 turns each and carries current $I_0 = 5.57 \text{ A}$. Find B at centre.

Sol.

DATA

$$\ell = 1.23 \text{ m}$$

$$d = 3.55 \text{ cm}$$

$$I_0 = 5.57 \text{ A}$$

$$B = ?$$

$$N = 5 \times 850 = 4250$$

CALCULATIONS

$$B = \mu_0 n I_0 = \frac{\mu_0 N I_0}{\ell}$$

$$B = \frac{4\pi \times 10^{-7} \times 4250 \times 5.57}{1.23} = 0.24 \text{ T}$$

EXERCISE PROBLEMS

P:1 A copper wire having 2.6 mm diameter carries current of 50 A. what is magnetic field at the surface of wire?

sol

DATA $d =$

2.6 mm

270

$$r = 1.3 \text{ mm}$$

$$I = 50 \text{ A}$$

$$B = ? \quad 1.3 \text{ mm}$$



CALCULATIONS

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{4\pi \times 10^{-7} \times 50}{2\pi \times 1.3 \times 10^{-3}} = 7.6 \times 10^{-3} \text{ T}$$

- P:2 A surveyor is using a magnetic compass 6.3 m below a power line in which there is a steady current of 120 A. Will this interfere seriously with the compass reading. The horizontal component of earth magnetic field at the site is 21 μT .

Sol.

DATA

$$r = 6.3 \text{ m}$$

$$I = 120 \text{ A}$$

$$B = ?$$

CALCULATIONS

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{4\pi \times 10^{-7} \times 120}{2\pi \times 6.3} = 3.8 \mu\text{T}$$

The magnetic field due to current in power line is 3.8 μT . The magnetic field at the site due to earth is 21 μT . Therefore, it will interfere with earth magnetic field.

- P:3 The 25 kV electron gun tube fires an electron beam 0.22 mm in diameter at the screen, 5.6×10^{14} electrons arriving each second. Calculate the magnetic field produced by the beam at a point 1.5 mm from the axis of the beam.

Sol.

DATA

$$d = 0.22 \text{ mm}$$

$$r = 1.5 \text{ mm}$$

$$I = 5.6 \times 10^{14} \text{ electron/s}$$

$$I = 5.6 \times 10^{14} \times 1.6 \times 10^{-19} \text{ A} = 8.9 \times 10^{-5} \text{ A}$$

$$V = 25 \times 10^3 \text{ Volt}$$

$$B = ?$$

CALCULATIONS

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{4\pi \times 10^{-7} \times 8.9 \times 10^{-5}}{2\pi \times 1.5 \times 10^{-3}} = 1.1 \times 10^{-8} \text{ T}$$

P:4 At a location in Philippines,. the earth magnetic field has a value of $39 \mu\text{T}$ which is horizontal and due north. The net magnetic field is zero at 8.13 cm above a long straight horizontal wire that carries a steady Current. (a) Calculate current (b) direction of current

Sol.

DATA

$$B = 39 \mu\text{T}$$

$$r = 8.13 \text{ cm}$$

$$I = ?$$

CALCULATIONS

$$(a) \quad B = \frac{\mu_0 I}{2\pi r}$$

$$39 \times 10^{-6} = \frac{4\pi \times 10^{-7} \times I}{2\pi \times 8.13 \times 10^{-2}}$$

$$I = 16 \text{ A}$$

(b) The direction will be due north.

P:5 wooden A student cylinder makes of an diameter electromagnet 4.80 cm. by winding The coil 320 is turns connected of wire to a around battery.

producing a current of 4.2 A in the wire. (a) What is magnetic moment of this device? (b) At what axial distance $z \gg d$ will the B be $5 \mu\text{T}$.

Sol.

DATA

$$N = 320$$

$$d = 4.8 \text{ cm}$$

$$r = 2.4 \text{ cm}$$

$$I = 4.5 \text{ A}$$

$$B = 5 \mu\text{T}$$

$$\mu = ?$$

$$z = ?$$

CALCULATIONS

(a)

$$\mu = N I A$$

$$\mu = N I \pi r^2$$

$$\mu = 320 \times 4.2 \times \pi \times (2.4 \times 10^{-2})^2$$

$$\mu = 2.43 \text{ A-m}^2$$

(b)

$$B = \frac{\mu_0 \mu}{2\pi z^3}$$

$$z = \left(\frac{\mu_0 \mu}{2\pi B} \right)^{1/3}$$

$$z = \left(\frac{4\pi \times 10^{-7} \times 2.43}{2\pi \times 5 \times 10^{-6}} \right)^{1/3} = 0.45 \text{ m}$$

device? (b) At what axial distance z will the B be 5 PT .

PUACP

Consider a long cylindrical wire of radius R carrying a current I distributed uniformly over the cross-sectional area. At what two distances from the axis of the wire is B due to current equal to one-half the value at the surface?

sol.

The magnetic field at the surface of wire of radius R is

$$\frac{\mu_0 I}{2\pi R}$$

The magnetic field due to current I at distance r from center of wire is

$$\frac{\mu_0 I r}{2\pi R^2}$$

The given condition is

$$\frac{\mu_0 I}{2\pi R} = \frac{\mu_0 I r}{2\pi R^2}$$

A solenoid has length 1.33 m and diameter 2.6 cm and carries current of 17.8 A. The magnetic field inside the solenoid is 22.4 mT. Find the length of wire forming solenoid? Sol.

DATA

L 1.33 m
2.6 cm

B 22.4 mT

$I = 17.8$ A Length

of wire = ?

CALCULATIONS

$$B = \frac{\mu_0 N I}{L}$$

$$22.4 \times 10^{-3} = \frac{4\pi \times 10^{-7} \times N \times 17.8}{1.33}$$

$N = 1332$ turns

One turn of wire has length $2\pi r$

1332 turns of wire has length — $1332 \times 2\pi r = 109$ m

A toroid having squarer cross-section and 5.2 cm on edge, and inner radius 16.2 cm has 535 turns and carries a current of 813 mA. Calculate the B inside a toroid, at the inner and outer radius of toroid?

Sol.

$$\text{DATA at } a = 16.2 \text{ cm } b = 16.2 + 5.2 =$$

$$21.4 \text{ cm}$$

$$813 \text{ mA}$$

$$N = 535$$

$$= 18.8 \text{ cm}$$

$$2$$

CALCULATIONS

Magnetic field inside toroid

$$\frac{N}{2\pi r} \times 10^{-7} \times 535 \times 813 \times 10^{-3} = 10^{-4} \text{ T}$$

$$18.8 \times 10^{-2}$$

Magnetic field at inner radius $r = a$

$$16.2$$

$$\frac{535 \times 813 \times 10^{-3}}{2\pi \times 16.2 \times 10^{-2}} \times 10^{-7} = 10^{-4} \text{ T}$$

Magnetic field at

outer radius $r = b$

$$\frac{535 \times 813 \times 10^{-3}}{2\pi \times 21.4 \times 10^{-2}} \times 10^{-7} = 4.05 \times 10^{-5} \text{ T}$$

P:9 A solenoid has 100 turns per centimeter. An electron moves within the solenoid in a circle of radius 2.3 cm perpendicular to axis of solenoid. If speed of electron is 0.046 c. Find the current. Sol,

DATA

$$100 \text{ turns/cm}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$9.1 \times 10^{-31} \text{ kg}$$

$$2.3 \text{ cm}$$

$$v = 0.046 c$$

$$1$$

CALCULATIONS

$$F = \frac{m q V B}{r}$$

qr

The magnetic field inside a solenoid

Comparing eq(1) and eq(2)

9] AMPERE'S LAW

269

$$I = \frac{m V}{\mu_0 n q r}$$

$$I = \frac{m V}{\mu_0 n q r}$$

$$I = \frac{9.1 \times 10^{-31} \times 0.046 \times 3 \times 10^8}{4\pi \times 10^{-7} \times 10000 \times 1.6 \times 10^{-19} \times 2.3 \times 10^{-2}}$$

$$I = 0.27 \text{ A}$$



p:10 A solenoid has 95.6 cm long has a radius 1.9 cm, and winding of 1230 turns, carries a current of 3.58 A. Calculate the strength of magnetic field inside the solenoid.

Sol.

DATA

$$\ell = 95.6 \text{ cm}$$

$$r = 1.9 \text{ cm}$$

$$I_0 = 3.58 \text{ A}$$

$$B = ?$$

$$N = 1230$$

CALCULATIONS

$$B = \mu_0 n I_0 = \frac{\mu_0 N I_0}{\ell}$$

$$B = \frac{4\pi \times 10^{-7} \times 1230 \times 3.58}{95.6 \times 10^{-2}} = 5.78 \times 10^{-3} \text{ T}$$

P:11 A solenoid is 1.5 m long carries a current of 18 A. The magnetic field inside it is 22.5 mT. Find the total number of turns of the solenoid.

Sol.

DATA

$$\ell = 1.5 \text{ m}$$

$$I_0 = 18 \text{ A}$$

$$B = 22.5 \text{ mT}$$

$$N = ?$$

CALCULATIONS

$$B = \mu_0 n I_0 = \frac{\mu_0 N I_0}{\ell}$$

$$N = \frac{B \ell}{\mu_0 I_0}$$

$$N = \frac{22.5 \times 10^{-3} \times 1.5}{4\pi \times 10^{-7} \times 18}$$

$$N = 1493$$

carries a current of 3.58 A. Calculate the strength of magnetic field inside the

P:12 Find the value of magnetic field at distance 2 cm from a long straight conductor in which current 2mA flows.

Sol.

DATA

$$r = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$$

$$I = 2 \text{ mA} = 2 \times 10^{-3} \text{ A}$$

$$B = ?$$

CALCULATIONS

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{4\pi \times 10^{-7} \times 2 \times 10^{-3}}{2\pi \times 2 \times 10^{-2}} = 2 \times 10^{-7} \text{ T}$$

P:13 A solenoid of certain diameter is 3.14 m long. It has 5 layers and 850 turns each and carries a current 5 A. What is B at the center.

Sol.

DATA

$$\ell = 3.14 \text{ m}$$

$$I_0 = 5 \text{ A}$$

$$B = ?$$

$$N = 5 \times 850$$

CALCULATIONS

$$B = \mu_0 n I_0 = \frac{\mu_0 N I_0}{\ell}$$

$$B = \frac{4\pi \times 10^{-7} \times 5 \times 850 \times 5}{3.14}$$

$$B = 8.5 \times 10^{-3} \text{ T}$$

P:14 A 200 turns solenoid having a length of 25 cm carries a current of 0.30 A. Find the magnetic field B inside the solenoid.

Sol.

DATA

$$N = 200$$

$$\ell = 25 \text{ cm} = 0.25 \text{ m}$$

$$I_0 = 0.30 \text{ A}$$

$$B = ?$$

CALCULATIONS

$$B = \mu_0 n I_0 = \frac{\mu_0 N I_0}{\ell}$$

$$B = \frac{4\pi \times 10^{-7} \times 200 \times 0.30}{0.25}$$

$$B = 3.01 \times 10^{-4} \text{ T}$$

